Single-Qubit Quantum Gates: X, Y, Z, H

Quick Recap and Today's Question

A qubit is represented as a quantum state |psi> = alpha|0> + beta|1>. The Bloch Sphere helps visualise a single qubit, superposition allows a qubit to exist as a combination of basis states, and measurement converts a quantum state into a classical outcome, causing state collapse. Today's question is: if we know what a qubit is, how do we change its state? The answer is quantum gates.

Why Do We Need Quantum Gates

A quantum computer that cannot modify qubits cannot perform computations. Quantum gates are the building blocks that change the state of qubits, just as classical computers use logic gates such as AND, OR, and NOT.

Classical Gates vs Quantum Gates

Classical Gates	Quantum Gates
Operate on bits	Operate on qubits
Perform logical operations	Perform state transformations
Input is always 0 or 1	Input can be any valid quantum state
Example: NOT gate	Example: X Gate

Unlike classical gates, quantum gates are reversible: every gate has an inverse operation, allowing quantum computations to preserve information. Instead of thinking of gates as changing 0 to 1, think of them as rotating the qubit's state on the Bloch Sphere.

The Four Gates at a Glance

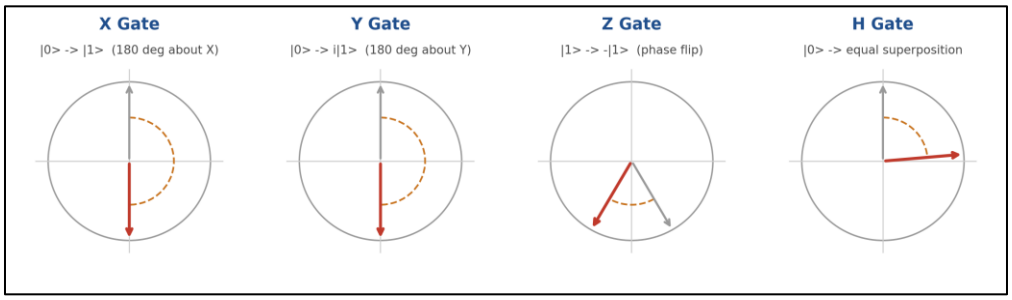


Figure 1: Each gate rotates the qubit's arrow to a different position on the Bloch Sphere.

1. X Gate (Quantum NOT Gate)

The X Gate is the quantum equivalent of the classical NOT gate. It swaps the basis states: |0> becomes |1>, and |1> becomes |0>. It rotates the qubit by 180 degrees about the X-axis. For basic states it behaves exactly like a NOT operation, but it works on any quantum state, not just basis states.

2. Y Gate

The Y Gate flips the qubit and changes its phase at the same time, which makes it slightly more complex than the X Gate. It rotates the qubit by 180 degrees about the Y-axis. Applying Y to |0> gives i|1>, where the i represents a phase change.

3. Z Gate

Unlike the X Gate, the Z Gate does not swap |0> and |1>. Instead it changes the phase of the |1> state: |0> stays |0>, while |1> becomes -|1>. It rotates the qubit by 180 degrees about the Z-axis. Although the measurement probabilities remain the same, the changed phase affects how the qubit behaves when combined with other gates, which is one of the reasons quantum algorithms can exploit interference.

4. Hadamard (H) Gate

The Hadamard Gate is one of the most important gates in quantum computing. Its primary purpose is to create equal superposition. Applying H to |0> gives (|0>+|1>)/root2, and applying H to |1> gives (|0>-|1>)/root2, a state with a different phase. Instead of simply flipping the qubit, the H Gate moves the state from the Z-axis toward the X-axis. Many famous quantum algorithms begin by applying the H Gate, including Grover's Search Algorithm, the Deutsch-Jozsa Algorithm, Shor's Algorithm, and the Quantum Fourier Transform.

Comparing the Four Gates

Gate	Main Action	Bloch Rotation	Changes Phase
X	Flips basis states	180 deg about X-axis	No
Y	Flip + phase	180 deg about Y-axis	Yes
Z	Phase flip	180 deg about Z-axis	Yes
H	Creates superposition	Moves state toward X-basis	Yes

Common Beginner Mistakes

- "The X Gate is always the same as a NOT gate." In reality it behaves like NOT only for basis states; for arbitrary quantum states it transforms the amplitudes
- "The Z Gate does nothing because the measurement doesn't change." In reality the Z Gate changes the phase, which affects later operations
- "The H Gate gives both 0 and 1 as outputs." In reality it prepares the qubit in a superposition; a measurement still yields only one classical outcome

Summary: What to Remember

- Quantum gates change a qubit's state and are the building blocks of quantum computation
- Quantum gates are reversible, unlike most classical logic gates
- The X Gate flips basis states, rotating 180 degrees about the X-axis
- The Y Gate flips the qubit and changes its phase, rotating about the Y-axis
- The Z Gate leaves probabilities unchanged but flips the phase of |1>
- The Hadamard Gate creates equal superposition and starts many quantum algorithms
- Gates are best understood as rotations of the qubit's arrow on the Bloch Sphere